

Applied Mathematics for Chemists

(화학수학)

Lecture 19

Group Theory

기말고사

- 시간: 12월 15일(수) 오후 3시~4시
- 장소:
- 계산기 불필요, 신분증 지참
- 10분 전까지 입실 완료
- 시험 범위: 미분방정식(12장), 연산자, 행렬, 군론(13장)
- 수업 시간에 다루지 않은 내용은 범위에 포함되지 않음

Group (군)

Collection of members
with specific mathematical properties

Useful for studying systems with symmetry

Applications

Molecular orbitals of polyatomic molecules

Spectroscopic selection rules

Requirements for a Group

1. If A and B are members of the group, then the **product** AB is also a member of the group
2. The group must contain the **identity** E such that

$$AE = EA = A \text{ (for every member } A)$$

3. The **inverse** of every member A^{-1} is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds: $A(BC) = (AB)C$

Point Group (점군)

Formed by a set of point symmetry operators
belonging to a symmetrical object

Representation of a Point Group

Symmetry operator $\leftrightarrow$ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

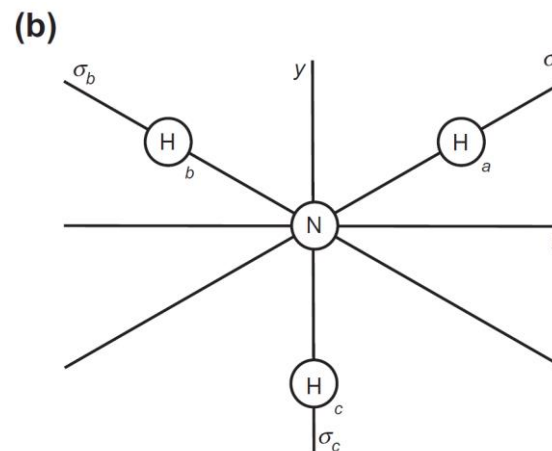
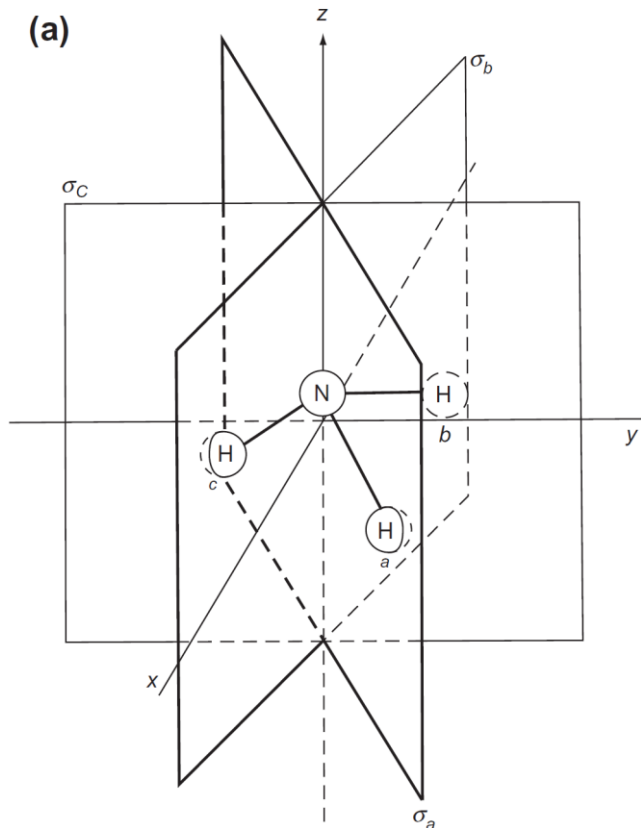
How to Find the Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{cases} a_{11}x_1 + a_{12}y_1 + a_{13}z_1 = x_2 \\ a_{21}x_1 + a_{22}y_1 + a_{23}z_1 = y_2 \\ a_{31}x_1 + a_{32}y_1 + a_{33}z_1 = z_2 \end{cases}$$

From the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) ,
determine a_{ij}

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

Similarity Transformation (답음 변환)

- Two different representations **P** and **Q** in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

when **X** is some square matrix

- Since $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr}(\mathbf{P}) = \text{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \text{tr}(\mathbf{XX}^{-1}\mathbf{Q}) = \text{tr}(\mathbf{Q})$$

Character (지표)

- **Character** of a symmetry operator = **trace** of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

$$\text{Character of } \hat{C}_3 = \text{tr } \mathbf{A} = (-1/2) + (-1/2) + 1 = 0$$

Our Representation of C_{3v} Group

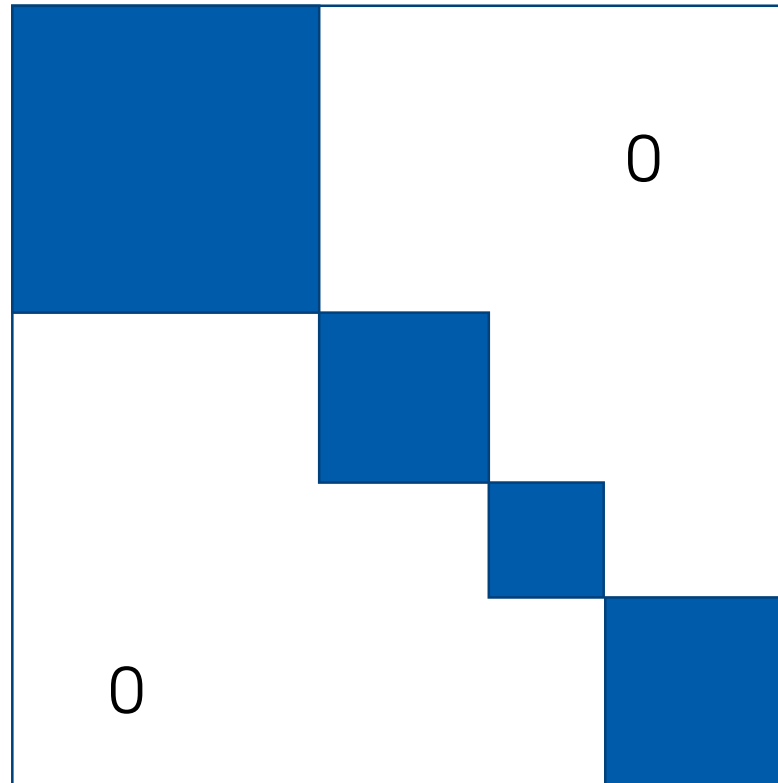
$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
3	0	0	1	1	1

Block-Diagonal Matrix (블록대각행렬)



Reducible Representation (가약 표현)

All matrices are **block-diagonal**, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{E}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{B}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & | & 0 \\ \sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & | & 0 \\ -\sqrt{3}/2 & -1/2 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{D}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & | & 0 \\ 0 & 1 & | & 0 \\ \hline 0 & 0 & | & 1 \end{bmatrix} = \mathbf{F}$$

Block Submatrices

- Block submatrices obey the same multiplication table of the group of symmetry operators
- Corresponding block submatrices can form a new representation

Example: 2 by 2 Blocks

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Example: 1 by 1 Blocks

$$\begin{aligned} \hat{E} &\leftrightarrow 1, & \hat{C}_3 &\leftrightarrow 1, & \hat{C}_3^2 &\leftrightarrow 1, \\ \hat{\sigma}_a &\leftrightarrow 1, & \hat{\sigma}_b &\leftrightarrow 1, & \hat{\sigma}_c &\leftrightarrow 1 \end{aligned}$$

	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$
$\hat{\sigma}_a$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_b$	$\hat{\sigma}_b$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_c$	$\hat{\sigma}_c$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal
by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as **irrep**

Examples of C_{3v}

- 2 by 2 blocks: one irrep

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}, \quad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- 1 by 1 blocks: another irrep

$$\hat{E} \leftrightarrow 1, \quad \hat{C}_3 \leftrightarrow 1, \quad \hat{C}_3^2 \leftrightarrow 1, \quad \hat{\sigma}_a \leftrightarrow 1, \quad \hat{\sigma}_b \leftrightarrow 1, \quad \hat{\sigma}_c \leftrightarrow 1$$

Characters of Irreps

Class:	E	$2C_3$		$3\sigma_v$		
Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
2 by 2	2	-1	-1	0	0	0
1 by 1	1	1	1	1	1	1

Class is constituted of symmetry operations
that can be converted by similarity transformations

Character Table (지표표)

- Using group theory theorems, we can find all possible irreps and collect their characters → **character table**
- e.g.* C_{3v} group

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

conventional notation for irreps

Review:

Differential Equations

Operators and Matrices

12. Differential Equations

(12장 미분방정식)

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$

Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions

(일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order n ,

- **General solution:** contains n arbitrary constants
- **Particular solution:** contains no arbitrary constants
 - Particular solutions can be generated from the general solution



Additional Conditions

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n - 1)$ derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
 2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
 - For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**

Exact Differential Equations

- Pfaffian form:

$$M(x, y) \, dx + N(x, y) \, dy = 0$$

If the differential equation is **exact**,
there is a function $f(x, y)$ such that

$$df = M(x, y) \, dx + N(x, y) \, dy = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $f(x, y)$

$$df = M(x, y) dx + N(x, y) dy = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C df$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1) . Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) dx + \int_{y_0}^{y_1} N(x_1, y) dy$$

This gives the relationship between x and y .

Integrating Factors

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **inexact**, but we know the **integration factor** $g(x, y)$ that makes

$$g(x, y)M(x, y) dx + g(x, y)N(x, y) dy = 0$$

an exact differential equation, we can solve this DE.

Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + C e^{-\int P(x)dx}$$

Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs

Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.

Reduction of Order

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain the other solution y_2

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Three possible cases!

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Guess for Particular Solution

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos(at)$
$\frac{a}{s^2 + a^2}$	$\sin(at)$
$\frac{s}{s^2 - a^2}$	$\cosh(at)$
$\frac{a}{s^2 - a^2}$	$\sinh(at)$

Useful Theorems

- Shifting Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

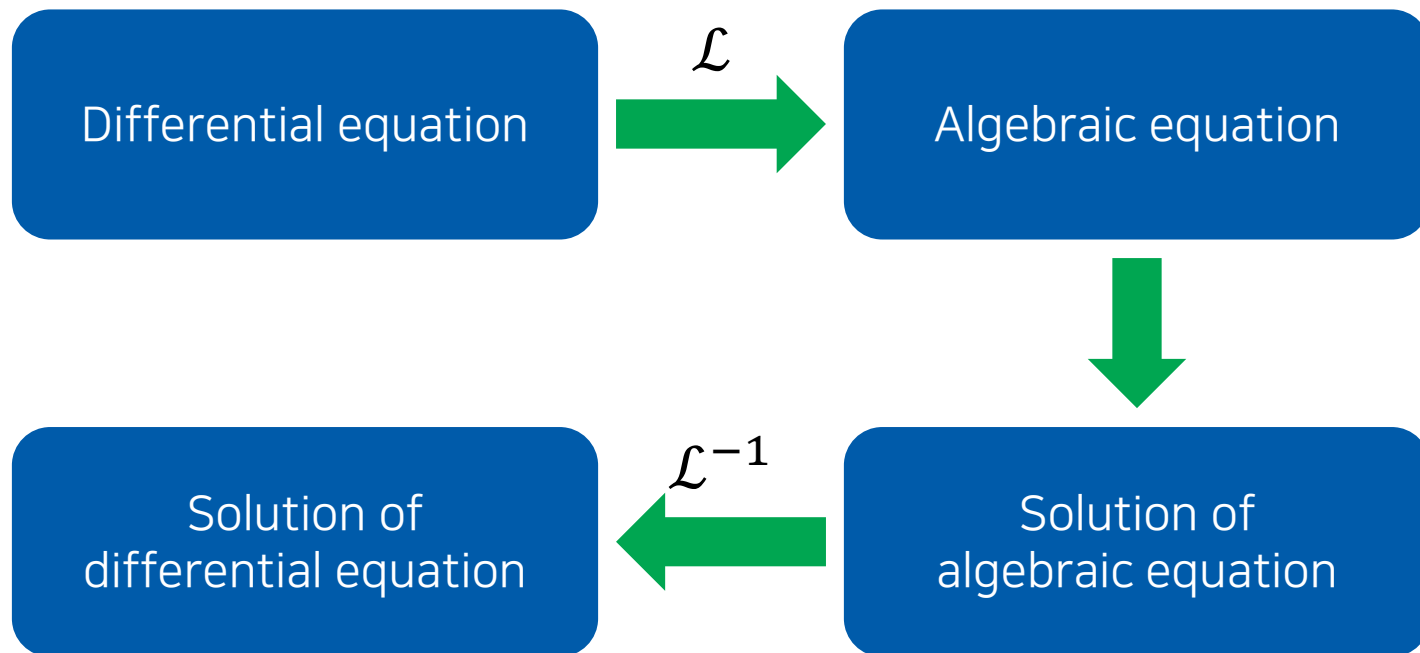
- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$

Solving DE with Laplace Transforms



Partial Differential Equations

- General form of a **linear second-order PDE**

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \dots, G$ are functions of x and y .

- $G(x, y) = 0$: **homogeneous**
- $G(x, y) \neq 0$: **inhomogeneous**

Separation of Variables

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Assume that $u(x, y) = X(x)Y(y)$.

$$\begin{aligned} X' &= dX/dx, X'' = d^2X/dx^2, \dots \\ Y' &= dY/dy, Y'' = d^2Y/dy^2, \dots \end{aligned}$$

If we obtain $f(x, X, X', \dots) = g(y, Y, Y', \dots)$, let

$$f(x, X, X', \dots) = g(y, Y, Y', \dots) = \lambda$$

where λ is the **separation constant**.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

13. Operators, Matrices, and Group Theory

(13장 연산자, 행렬, 군론)

Mathematical Operators (연산자)

perform **mathematical operations** on some function

e.g. "+3" operator, "×2" operator, "d/dx" operator

- Typically denoted by a letter with a **caret** (^)
- Operate on the function on its right

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A} ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator** (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A} \cdots \hat{A} \text{ (} n \text{ times)}$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$
- Not all operators have inverses (*e.g.* $\hat{0}$)

Operator Algebra and Differential Equation

- Derivative operator $\hat{D}_x = d/dx$
- Homogeneous linear DE with constant coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

$$(a_n \hat{D}_x^n + a_{n-1} \hat{D}_x^{n-1} + \cdots + a_1 \hat{D}_x + a_0 \hat{E})y = 0$$

This can be solved by operator algebra

Eigenfunctions and Eigenvalues

(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an **eigenfunction** of $\hat{A}$, and a is called an **eigenvalue**. This equation is called an **eigenvalue equation** (고윳값 방정식).

Operators in Quantum Mechanics

“For every **observable quantity**,
there is a corresponding **mathematical operator**”

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z)$$

Hermitian Property (에르미트 성질)

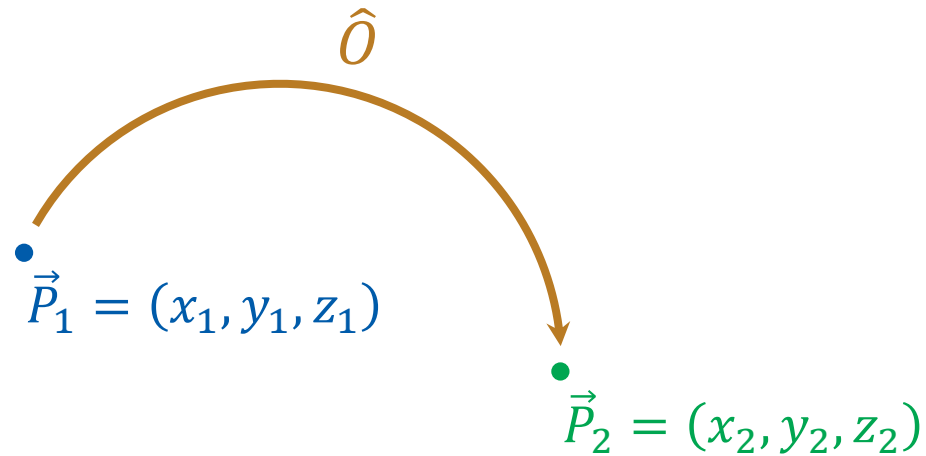
- If $\hat{A}$ is **Hermitian**, it obeys the relation

$$\int \chi^* (\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators

(점대칭 연산자)

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator $\hat{E}$

Inversion operator $\hat{i}$

Reflection operator $\hat{\sigma}$

Rotation operator $\hat{C}_n, \hat{S}_n$

Symmetry Elements (대칭 요소)

Point, line, or plane relative to which the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol
- Inversion i : the origin
- Reflection σ : the reflection plane
- Rotation C_n, S_n : the rotation axis

Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, **the symmetry operator belongs to the object.**
- Coordinate system
 - origin: center of mass of the object
 - z axis: the rotation axis of highest order (largest n)

Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$,
for a function f ,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) !

Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

the operator leaves a function unchanged

- Eigenvalue = -1

the operator changes the sign of the function's value at all points

Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

matrix elements
(행렬 원소)

(m by n matrix)

Matrix Algebra

- Two matrices are **equal** if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix

- **Sum of two matrices:** $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- **Product of a scalar c and a matrix $\mathbf{A}$:** $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

where n is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

$$(m \text{ by } n \text{ matrix}) \times (n \text{ by } p \text{ matrix}) = (m \text{ by } p \text{ matrix})$$

Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$

Full Bag of New Terms

1. **Special matrices:** identity matrix, zero matrix
2. **Partner of a matrix:** inverse, transpose, Hermitian conjugate
3. **Properties of a matrix:** singular, symmetric, Hermitian, orthogonal, unitary
4. **Derived quantities from a matrix:** trace, determinant

Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^n a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

Gauss-Jordan Elimination

- Construct an **augmented matrix** (첨가행렬)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array} \right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array} \right] A^{-1}$$

Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Jacobian calculation
- Determinant of a singular matrix
- Matrix eigenvalue problem

Matrix Eigenvalue Problem

For a given square matrix $\mathbf{B}$, if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$ (column vector): eigenvector
- b (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$